

Quend 700

HIGH YIELD STRENGTH STEEL

QUEND 700 is extra high strength structural steel produced as quenched and tempered with a minimum yield strength of 700 MPa. **QUEND 700** complies with the requirements corresponding to the S690QL given in the EN 10025-6 standard, where a minimum impact toughness of 27 J is guaranteed at -40°C.

1. Applications

QUEND 700 is recommended for the following applications:

- truck chassis
- lifting and hoisting equipment
- handling equipment
- trailers

2. Technical characteristics

Tensile properties

TRANSVERSE TESTING		
Yield strength Rp 0.2	Tensile Strength Rm	Elongation A5
700 MPa min	780 - 930 MPa	14% min

Impact toughness

MINIMUM VALUES AT		
0 °C	-20 °C	-40 °C
35 J	30 J	27 J

Transverse testing according to EN 10025 option 30. Thickness < 12 mm subsize Charpy V specimen have been used.

Testing according to EN 10025.

Chemical composition

The steel is fine grain treated.

MAX LADLE ANALYSIS, %												
C	Si	Mn	P	S	Nb	Cr	V	Ti	Ni	Al	Mo	N
0,20	0,60	1,50	0,020	0,010	0,040	0,80	0,070	0,040	1,00	0,070	0,50	0,014

CARBON EQUIVALENT, TYPICAL VALUES, %		
Thickness	CEV ⁽¹⁾	CET ⁽²⁾
4 - 15 mm	0.45	0.29
15.01 - 25 mm	0.44	0.30
25.01 - 40 mm	0.45	0.30
40.01 - 50 mm	0.54	0.33

(1) CEV = C+Mn/6+ (Ni+Cu)/15+ (Cr+Mo+V)/5, (2) CET = C+(Mn+Mo)/10+Ni/40 +(Cr+Cu)/20

3. Dimensions

QUEND 700 is currently supplied in the following range:

- thickness: 4 - 50 mm
- width: 1500 - 3100 mm

For more information, please check our website or contact your local NLMK Clabecq representative.

4. Flatness, tolerances & surface properties

QUEND 700 is delivered with a unique combination of excellent flatness, tight thickness tolerances and superior surface finish.

FEATURE	NORM
FLATNESS	- EN 10029: - Class N (standard) & - Class S
THICKNESS tolerance	- meets and exceeds EN 10029 Class A - tighter tolerances upon request
Shape, length, width tolerances	meets EN 10029
SURFACE properties	exceeds the usual market standards, EN 10163-2 Class B3

5. Delivery conditions

QUEND 700 is delivered as quenched and tempered. Our QUEND plates are supplied as standard in the shotblasted and painted condition. In order to maintain a good weldability and laser cutting performance, a low zinc silicate primer is applied. Plates can also be delivered unpainted.

6. Heat treatment

The mechanical properties of QUEND 700 has been obtained by quenching and tempering. For not losing the guaranteed properties of QUEND 700, the plate should not be used in applications requiring hot working and service temperatures above 550 °C.

7. Ultrasonic testing

Ultra sonic testing (UT), is applied to secure the plate from discontinuities like inclusions, cracks and porosity. In thickness from 8 mm and up, all plates are UT tested and controlled against class S2, E2, according to EN 10160.

8. General processing recommendations

To obtain optimal work shop productivity when processing QUEND 700, it is essential to use the recommended procedures and tools given below.

Thermal cutting

QUEND 700 may be cut either by oxygen fuel, plasma and laser cutting without any restrictions. Subsequent to cutting, let the cut parts slowly cool down to room temperature. Do never accelerate the cooling of the parts. A slow cooling rate will reduce the risk of cut edge cracking.

Cold forming

QUEND 700 is very well suited for cold forming operations. QUEND 700 complies with the S690QL bending requirements but offer even closer R/t ratios:

Thickness (mm)	Transverse to rolling (R/t)	Longitudinal to rolling (R/t)	Trans. Width (W/t)	Long. Width (W/t)
$t < 8.0$	1.5	2.0	8	9
$8 \leq t < 20$	2.0	3.0	8	9
$t \geq 20.0$	3.0	4.0	9	10

R = Recommended punch radius (mm), t = Plate thickness (mm), W – Die opening width (mm) (bending angle $\leq 90^\circ$)

Due to the homogeneous properties and narrow thickness tolerances of QUEND 700, variations in springback are kept at a low level. Grinding of flame cut or a sheared edge in the bending area is recommended to further prevent cracking during bending.

Welding

Welding of QUEND 700 can be performed using any of the conventional welding methods available both as manual or automatic.

In the thickness range up to 30 mm, preheating prior to welding is normally not needed, if a heat input of 1,7 kJ/mm is used. Welding of QUEND 700 is recommended to be performed at ambient temperature not lower than +5 °C. Subsequent to welding, let the welded parts slowly cool down to room temperature. Do never accelerate the cooling process of the weld.

It is always recommended to use low hydrogen electrodes when welding QUEND 700.

Machining

QUEND 700 provides a very good machinability and can be drilled, counter sunked and milled in the same way as any other 700MPa or S690QL Q&T steel.



If you want to calculate the optimal parameters for your operations on QUARD and QUEND, download our Q Calculator using the QR code. Available on Google Play and Apple Store.



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Quend 900

HIGH YIELD STRENGTH STEEL

QUEND 900 is extra high strength structural steel produced as quenched and tempered with a minimum yield strength of 900 MPa. **QUEND 900** complies with the requirements corresponding to the S890QL given in the EN 10025-6 standard, where a minimum impact toughness of 27 J is guaranteed at -40°C

1. Applications

QUEND 900 is recommended for the following applications:

- crane booms
- lifting equipment
- stabilising support
- undercarriage

2. Technical characteristics

Tensile properties

TRANSVERSE TESTING		
Yield strength Rp 0.2	Tensile Strength Rm	Elongation A5
900 MPa min	940 - 1100 MPa	12% min

Impact toughness

MINIMUM VALUES AT		
0 °C	-20 °C	-40 °C
35 J	30 J	27 J

Transverse testing according to EN 10025 option 30. Thickness < 12 mm subsize Charpy V specimen have been used.

Testing according to EN 10025.

Chemical composition

The steel is fine grain treated

MAX LADLE ANALYSIS, %													
C	Si	Mn	P	S	Nb	Cr	V	Ti	Ni	Al	Mo	N	B
0,20	0,50	1,50	0,02	0,01	0,04	0,70	0,06	0,01	1,50	0,06	0,70	0,005	0,005

CARBON EQUIVALENT, TYPICAL VALUES, %	
CEV(1)	CET(2)
0,57	0,36

(1) CEV = C+Mn/6+ (Ni+Cu)/15+ (Cr+Mo+V)/5, (2) CET = C+(Mn+Mo)/10+Ni/40 +(Cr+Cu)/20

3. Dimensions

QUEND 900 is currently supplied in the following range:

- thickness: 4 - 40 mm
- width: 1500 - 3100 mm

NLMK Clabecq carries on the extension of its dimensional program. For more information, please check our website or contact your local representative.

4. Flatness, tolerances & surface properties

QUEND 900 is delivered with a unique combination of excellent flatness, tight thickness tolerances and superior surface finish.

FEATURE	NORM
FLATNESS	- EN 10029: - Class N (standard) & - Class S
THICKNESS tolerance	- meets and exceeds EN 10029 Class A - tighter tolerances upon request
Shape, length, width tolerances	meets EN 10029
SURFACE properties	exceeds the usual market standards, EN 10163-2 Class B3

5. Delivery conditions

QUEND 900 is delivered as quenched and tempered. Our QUEND plates are supplied as standard in the shotblasted and painted condition. In order to maintain a good weldability and laser cutting performance, a low zinc silicate primer is applied. Plates can also be delivered unpainted.

6. Heat treatment

The mechanical properties of QUEND 900 has been obtained by quenching and tempering. For not losing the guaranteed properties of QUEND 900, the plate should not be used in applications requiring hot working and service temperatures above 550 °C.

7. Ultrasonic testing

Ultra sonic testing (UT), is applied to secure the plate from discontinuities like inclusions, cracks and porosity. In thickness from 8 mm and up, all plates are UT tested and controlled against class S2, E2, according to EN 10160.

8. General processing recommendations

To obtain optimal work shop productivity when processing QUEND 900, it is essential to use the recommended procedures and tools given below.

Thermal cutting

QUEND 900 may be cut either by oxygen fuel, plasma and laser cutting without any restrictions. Subsequent to cutting, let the cut parts slowly cool down to room temperature. Do never accelerate the cooling of the parts. A slow cooling rate will reduce the risk of cut edge cracking.

Cold forming

QUEND 900 is very well suited for cold forming operations. QUEND 900 complies with the S890QL bending requirements but offer even closer R/t ratios:

Thickness (mm)	Transverse to rolling (R/t)	Longitudinal to rolling (R/t)	Trans. Width (W/t)	Long. Width (W/t)
$t < 8.0$	2.5	3.0	9	10
$8 \leq t < 20$	3.0	4.0	9	10
$t \geq 20$	4.0	5.0	10	12

R = Recommended punch radius (mm), t = Plate thickness (mm) , W – Die opening width (mm) (bending angle $\leq 90^\circ$)

Due to the homogeneous properties and narrow thickness tolerances of QUEND 900, variations in springback are kept at a low level. Grinding of flame cut or a sheared edge in the bending area is recommended to further prevent cracking during bending.

Welding

Welding of QUEND 900 can be performed using any of the conventional welding methods available both as manual or automatic.

In the thickness range up to 10 mm, preheating prior to welding is normally not needed, if a heat input of 1,7 kJ/mm is used. Welding of QUEND 900 is recommended to be performed at ambient temperature not lower than +5 °C. Subsequent to welding, let the welded parts slowly cool down to room temperature. Do never accelerate the cooling process of the weld.

It is always recommended to use low hydrogen electrodes when welding QUEND 900.

Machining

QUEND 900 provides a very good machinability and can be drilled, counter sunked and milled in the same way as any other 900MPa or S890QL Q&T steel.



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Quend 960

HIGH YIELD STRENGTH STEEL

QUEND 960 is extra high strength structural steel produced as quenched and tempered with a minimum yield strength of 960 MPa. **QUEND 960** complies with the requirements corresponding to the S960QL given in the EN 10025-6 standard, where a minimum impact toughness of 27 J is guaranteed at -40°C

1. Applications

QUEND 960 is recommended for the following applications:

- crane booms
- lifting equipment
- stabilising support
- undercarriage

2. Technical characteristics

Tensile properties

TRANSVERSE TESTING		
Yield strength Rp 0.2	Tensile Strength Rm	Elongation A5
960 MPa min	980 - 1150 MPa	12% min

Impact toughness

MINIMUM VALUES AT		
0 °C	-20 °C	-40 °C
35 J	30 J	27 J

Transverse testing according to EN 10025 option 30. Thickness < 12 mm subsized charpy V specimen have been used.

Testing according to EN 10025.

Chemical composition

The steel is fine grain treated.

MAX LADLE ANALYSIS, %													
C	Si	Mn	P	S	Nb	Cr	V	Ti	Ni	Al	Mo	N	B
0,20	0,50	1,50	0,02	0,01	0,04	0,70	0,06	0,01	1,50	0,06	0,70	0,005	0,005

CARBON EQUIVALENT, TYPICAL VALUES, %	
CEV(1)	CET(2)
0,57	0,36

(1) CEV = C+Mn/6+ (Ni+Cu)/15+ (Cr+Mo+V)/5, (2) CET = C+(Mn+Mo)/10+Ni/40 +(Cr+Cu)/20

3. Dimensions

QUEND 960 is currently supplied in the following range:

- thickness: 4 - 35 mm
- width: 1500 - 3100 mm

NLMK Clabecq is continuing the extension of its dimensional program. For more information, please check our website or contact your local representative.

4. Flatness, tolerances & surface properties

QUEND 960 is delivered with a unique combination of excellent flatness, tight thickness tolerances and superior surface finish.

FEATURE	NORM
FLATNESS	- EN 10029: - Class N (standard) & - Class S
THICKNESS tolerance	- meets and exceeds EN 10029 Class A - tighter tolerances upon request
Shape, length, width tolerances	meets EN 10029
SURFACE properties	exceeds the usual market standards, EN 10163-2 Class B3

5. Delivery conditions

QUEND 960 is delivered as quenched and tempered. Our QUEND plates are supplied as standard in the shotblasted and painted condition. In order to maintain a good weldability and laser cutting performance, a low zinc silicate primer is applied. Plates can also be delivered unpainted.

6. Heat treatment

The mechanical properties of QUEND 960 has been obtained by quenching and tempering. For not losing the guaranteed properties of QUEND 960, the plate should not be used in applications requiring hot working and service temperatures above 550 °C.

7. Ultrasonic testing

Ultra sonic testing (UT), is applied to secure the plate from discontinuities like inclusions, cracks and porosity. In thickness from 8 mm and up, all plates are UT tested and controlled against class S2, E2, according to EN 10160.

8. General processing recommendations

To obtain optimal work shop productivity when processing QUEND 960, it is essential to use the recommended procedures and tools given below.

Thermal cutting

QUEND 960 may be cut either by oxygen fuel, plasma and laser cutting without any restrictions. Subsequent to cutting, let the cut parts slowly cool down to room temperature. Do never accelerate the cooling of the parts. A slow cooling rate will reduce the risk of cut edge cracking.

Cold forming

QUEND 960 is very well suited for cold forming operations. QUEND 960 complies with the S960QL bending requirements but offer even closer R/t ratios:

Thickness (mm)	Transverse to rolling (R/t)	Longitudinal to rolling (R/t)	Trans. Width (W/t)	Long. Width (W/t)
$t < 8.0$	2.5	3.0	9	10
$8 \leq t < 20$	3.0	4.0	9	10
$t \geq 20$	4.0	5.0	10	12

R = Recommended punch radius (mm), t = Plate thickness (mm), W – Die opening width (mm) (bending angle $\leq 90^\circ$)

Due to the homogeneous properties and narrow thickness tolerances of QUEND 960, variations in springback are kept at a low level. Grinding of flame cut or a sheared edge in the bending area is recommended to further prevent cracking during bending.

Welding

Welding of QUEND 960 can be performed using any of the conventional welding methods available both as manual or automatic.

In the thickness range up to 10 mm, preheating prior to weldin is normally not needed, if a heat input of 1,7 kJ/mm is used. Welding of QUEND 960 is recommended to be performed at ambient temperature not lower than +5 °C. Subsequent to welding, let the welded parts slowly cool down to room temperature. Do never accelerate the cooling process of the weld.

It is always recommended to use low hydrogen electrodes when welding QUEND 960.

Machining

QUEND 960 provides a very good machinability and can be drilled, counter sunk and milled in the same way as any other 960MPa or S960QL Q&T steel.



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Quend 1100

HIGH YIELD STRENGTH STEEL

QUEND 1100 is extra high strength structural steel produced as quenched and tempered with a minimum yield strength of 1100 MPa.

1. Applications

QUEND 1100 is recommended for the applications where high load bearing capacity is required:

- lifting equipment (mobile cranes, marine cranes, truck cranes)
- stabilising support, stabilising structures

2. Technical characteristics

Tensile properties

TRANSVERSE TESTING			
Yield strength Rp 0.2	Tensile Strength Rm	Thickness mm	Elongation A5
1100 MPa min	1250 - 1500 MPa	≤ 10mm > 10 mm	8% min 10% min

Impact toughness

MINIMUM VALUES AT		
0 °C	-20 °C	-40 °C
35 J	30 J	27 J

Transverse testing according to EN 10025 option 30. Thickness < 12 mm subsize Charpy V specimen have been used.

Testing according to EN 10025.

Chemical composition

The steel is fine grain treated.

MAX LADLE ANALYSIS, %													
C	Si	Mn	P	S	Nb	Cr	V	Ti	Ni	Al	Mo	N	B
0,20	0,50	1,50	0,02	0,01	0,04	0,70	0,06	0,01	1,50	0,06	0,70	0,005	0,005

CARBON EQUIVALENT, TYPICAL VALUES, %	
CEV(1)	CET(2)
0,57	0,36

(1) CEV = C+Mn/6+ (Ni+Cu)/15+ (Cr+Mo+V)/5, (2) CET = C+(Mn+Mo)/10+Ni/40 +(Cr+Cu)/20

3. Dimensions

QUEND 1100 is currently supplied in the following range:

- thickness: 4 - 12 mm
- width: 1500 - 3100 mm

NLMK Clabecq carries on the extension of its dimensional program in order to propose as soon as possible a thickness range from 4 to 40 mm. For more information, please check our website or contact your local representative.

4. Flatness, tolerances & surface properties

QUEND 1100 is delivered with a unique combination of excellent flatness, tight thickness tolerances and superior surface finish.

FEATURE	NORM
FLATNESS	- EN 10029: - Class N (standard) & - Class S
THICKNESS tolerance	- meets and exceeds EN 10029 Class A - tighter tolerances upon request
Shape, length, width tolerances	meets EN 10029
SURFACE properties	exceeds the usual market standards, EN 10163-2 Class B3

5. Delivery conditions

QUEND 1100 is delivered as quenched and tempered. Our QUEND plates are supplied as standard in the shotblasted and v-painted condition. In order to maintain a good weldability and laser cutting performance, a low zinc silicate primer is applied. Plates can also be delivered unpainted.

6. Heat treatment

The mechanical properties of QUEND 1100 has been obtained by quenching and tempering. For not losing the guaranteed properties of QUEND 1100, the plate should not be used in applications requiring hot working and service temperatures above 200 °C.

7. Ultrasonic testing

Ultra sonic testing (UT), is applied to secure the plate from discontinuities like inclusions, cracks and porosity. In thickness from 8 mm and up, all plates are UT tested and controlled against class S2, E2, according to EN 10160.

8. General processing recommendations

To obtain optimal work shop productivity when processing QUEND 1100, it is essential to use the recommended procedures and tools given below.

Thermal cutting

QUEND 1100 may be cut either by oxygen fuel, plasma and laser cutting without any restrictions. Subsequent to cutting, let the cut parts slowly cool down to room temperature. Do never accelerate the cooling of the parts. A slow cooling rate will reduce the risk of cut edge cracking.

Cold forming

QUEND 1100 is very well suited for cold forming operations. QUEND 1100 complies with the S1100QL bending requirements but offer even closer R/t ratios:

Thickness (mm)	Transverse to rolling (R/t)	Longitudinal to rolling (R/t)	Trans. Width (W/t)	Long. Width (W/t)
$t < 8.0$	3.0	3.5	10	10
$8 \leq t < 20$	3.0	3.5	10	12

R = Recommended punch radius (mm), t = Plate thickness (mm), W – Die opening width (mm) (bending angle $\leq 90^\circ$)

Due to the homogeneous properties and narrow thickness tolerances of QUEND 1100, variations in springback are kept at a low level. Grinding of flame cut or a sheared edge in the bending area is recommended to further prevent cracking during bending.

Welding

Welding of QUEND 1100 can be performed using any of the conventional welding methods available both as manual or automatic.

In the thickness range up to 10 mm, preheating prior to welding is normally not needed, if a heat input of 1,7 kJ/mm is used. Welding of QUEND 1100 is recommended to be performed at ambient temperature not lower than +5 °C. Subsequent to welding, let the welded parts slowly cool down to room temperature. Do never accelerate the cooling process of the weld.

It is always recommended to use low hydrogen electrodes when welding QUEND 1100.

Machining

QUEND 1100 provides a very good machinability and can be drilled, counter sunk and milled in the same way as any other 1100MPa Q&T steel.



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Quend 1300

HIGH YIELD STRENGTH STEEL

QUEND 1300 is extra high strength structural steel produced as quenched and tempered with a minimum yield strength of 1300 MPa.

1. Applications

QUEND 1300 is recommended for the applications where high load-bearing capacity is required:

- lifting equipment (mobile cranes, marine cranes, truck cranes)
- stabilising support, stabilising structures

2. Technical characteristics

Tensile properties

TRANSVERSE TESTING		
Yield strength Rp 0.2	Tensile Strength Rm	Elongation A5
1300 MPa min	1400 - 1700 MPa	8% min

Impact toughness

MINIMUM VALUES AT		
0 °C	-20 °C	-40 °C
35 J	30 J	27 J

Transverse testing according to EN 10025 option 30. Thickness < 12 mm subsize Charpy V specimen have been used.

Testing according to EN 10025.

Chemical composition

The steel is fine grain treated.

MAX LADLE ANALYSIS, %													
C	Si	Mn	P	S	Nb	Cr	V	Ti	Ni	Al	Mo	N	B
0,25	0,50	1,20	0,02	0,01	0,04	0,70	0,06	0,01	1,50	0,06	0,70	0,005	0,005

CARBON EQUIVALENT, TYPICAL VALUES, %	
CEV(1)	CET(2)
0,67	0,43

(1) CEV = C+Mn/6+ (Ni+Cu)/15+ (Cr+Mo+V)/5, (2) CET = C+(Mn+Mo)/10+Ni/40 +(Cr+Cu)/20

3. Dimensions

QUEND 1300 is currently supplied in the following range:

- thickness: 4 - 12 mm
- width: 1500 - 2700 mm

NLMK Clabecq is continuing the extension of its dimensional program. For more information, please check our website or contact your local representative.

4. Flatness, tolerances & surface properties

QUEND 1300 is delivered with a unique combination of excellent flatness, tight thickness tolerances and superior surface finish.

FEATURE	NORM
FLATNESS	- EN 10029: - Class N (standard) & - Class S
THICKNESS tolerance	- meets and exceeds EN 10029 Class A - tighter tolerances upon request
Shape, length, width tolerances	meets EN 10029
SURFACE properties	exceeds the usual market standards, EN 10163-2 Class B3

5. Delivery conditions

QUEND 1300 is delivered as quenched and tempered. Our QUEND plates are supplied as standard in the shotblasted and painted condition. In order to maintain a good weldability and laser cutting performance, a low zinc silicate primer is applied. Plates can also be delivered unpainted.

6. Heat treatment

The mechanical properties of QUEND 1300 has been obtained by quenching and tempering. For not losing the guaranteed properties of QUEND 1300, the plate should not be used in applications requiring hot working and service temperatures above 200 °C.

7. Ultrasonic testing

Ultra sonic testing (UT), is applied to secure the plate from discontinuities like inclusions, cracks and porosity. In thickness from 8 mm and up, all plates are UT tested and controlled against class S2, E2, according to EN 10160.

8. General processing recommendations

To obtain optimal work shop productivity when processing QUEND 1300, it is essential to use the recommended procedures and tools given below.

Thermal cutting

QUEND 1300 may be cut either by oxygen fuel, plasma and laser cutting without any restrictions. Subsequent to cutting, let the cut parts slowly cool down to room temperature. Do never accelerate the cooling of the parts. A slow cooling rate will reduce the risk of cut edge cracking.

Cold forming

QUEND 1300 is very well suited for cold forming operations.

Thickness (mm)	Transverse to rolling (R/t)	Longitudinal to rolling (R/t)	Trans. Width (W/t)	Long. Width (W/t)
t < 8.0	3.5	4.0	10	10
8 ≤ t < 20	3.5	4.0	10	12

R = Recommended punch radius (mm), t = Plate thickness (mm) , W – Die opening width (mm) (bending angle ≤ 90°)

Due to the homogeneous properties and narrow thickness tolerances of QUEND 1300, variations in springback are kept at a low level. Grinding of flame cut or a sheared edge in the bending area is recommended to further prevent cracking during bending

Welding

Welding of QUEND 1300 can be performed using any of the conventional welding methods available both as manual or automatic.

In the thickness range up to 10 mm, preheating prior to welding is normally not needed, if a heat input of 1,7 kJ/mm is used. Welding of QUEND 1300 is recommended to be performed at ambient temperature, not lower than 5°C. For thickness ≥ 10 mm, preheating is recommended at 100°C. Subsequent to welding, let the welded parts slowly cool down to room temperature. Do never accelerate the cooling process of the weld.

It is always recommended to use low hydrogen electrodes when welding QUEND 1300.

Machining

QUEND 1300 provides a very good machinability and can be drilled, counter sunk and milled in the same way as any other 1300MPa Q&T steel.



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